FWD LM GCS – Pilot Operating Manual

Version: Working code (post Mission Planner parity and auto-apply mission updates)
Aircraft focus: Fixed-wing ArduPilot vehicles

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1. Purpose

This manual explains how to use the FWD LM Ground Control Station (GCS) for:

- Connecting to the vehicle
- Configuring video
- Calibrating the vehicle
- Planning missions
- Flying and monitoring the vehicle
- Using FWD LM custom in-flight override features

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2. System Overview

The FWD LM GCS is based on QGroundControl and customized for FWD fixed-wing operations. Important custom areas include:

- Mission Planner–style **Actions tab**
- **Quick tab** for live telemetry monitoring
- **Fly Here** map guidance
- **Auto-apply mission upload** during AUTO
- Compass calibration and compass priority support
- Maroon/black/white UI theme and fixed-wing iconography

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3. Connecting the Vehicle

3.1 USB / Serial Connection

1. Connect the vehicle or telemetry radio to the GCS computer.
2. Open FWD LM GCS.

3. Go to **Application Settings / Comm Links**.
4. Add a serial link:
5. Select the correct serial port.
6. Set baud rate to match the vehicle or telemetry radio.
7. Connect the link.
8. Wait for heartbeat and vehicle parameters to load.

3.2 UDP / Network Connection

1. Open **Comm Links**.
2. Add a UDP link.
3. Set the listening port, commonly `14550`.
4. Connect.
5. For SITL, use `127.0.0.1:14550`.

3.3 Connection Checks

Before flight, confirm:

- Vehicle appears in the vehicle list.
- Flight mode is displayed.
- Battery, GPS, and link status are valid.
- Parameters have loaded.
- No critical GCS or vehicle errors are shown.

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4. Video Connection

4.1 Configure Video Source

1. Open **Application Settings**.
2. Go to **Video**.
3. Select the video source:
4. UDP video stream
5. RTSP stream
6. File / test source, if configured
7. Enter the correct URL, port, or stream address.

4.2 Verify Video

1. Open **Fly View**.
2. Confirm the video feed appears.
3. Check latency and image quality.
4. Adjust resolution or bitrate at the transmitter if required.

4.3 Troubleshooting

- If video does not appear, verify the stream URL and port.
- Confirm the vehicle or camera is transmitting.
- Check firewall/network settings.
- Restart the video source if needed.

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5. Vehicle Calibration

5.1 Sensor Calibration

Open **Vehicle Setup → Sensors**.

Calibrate:

- Accelerometer
- Compass
- Gyroscope, if required
- Airspeed, if applicable
- Level horizon

5.2 Compass Calibration

The GCS supports:

- Normal onboard compass calibration
- Fast calibration for large vehicles

Normal Compass Calibration

1. Open **Sensors → Compass**.
2. Follow the on-screen rotation instructions.
3. Rotate the vehicle through all required axes.
4. Wait for completion and success message.

Fast Compass Calibration for Large Vehicles

Use this when moving the vehicle is difficult.

1. Point the vehicle North.
2. Open the compass calibration page.
3. Enable **Fast Calibration**.
4. Provide vehicle position if prompted.
5. Start calibration.

Note: Fast calibration assumes the magnetic field is close to spherical. It is not suitable for all installations.

5.3 Compass Priority

Where supported, compass priority can be configured using:

- `COMPASS_PRIO1_ID`
- `COMPASS_PRIO2_ID`
- `COMPASS_PRIO3_ID`

Use these parameters to prioritize the most reliable compass.

5.4 Other Calibrations

Also verify:

- Radio calibration
- Flight modes
- Failsafe settings
- Battery monitor
- Airspeed calibration, if applicable

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6. Plan View – Mission Planning

6.1 Open Plan View

1. Open **Plan View**.
2. The map and mission editor are displayed.
3. The home position is shown as the first mission item for ArduPilot.

6.2 Add Waypoints

1. Right-click on the map or use the mission editor.
2. Select **Add Waypoint** or desired mission command.
3. Move waypoints by dragging.
4. Edit altitude, speed, or command parameters in the mission item panel.

6.3 Mission Settings

Configure:

- Mission end action
- RTL behavior
- Loiter settings
- Camera or payload actions, if used

6.4 Save and Load Plans

- Save plans to file for reuse.
- Load existing plan files before flight.
- Verify the plan matches the intended mission.

6.5 Upload Mission

1. Click **Upload**.
2. Wait for upload completion.
3. Confirm no upload errors.

6.6 Auto-Apply Mission Upload During AUTO

FWD LM GCS supports uploading an updated mission while the aircraft is already flying in `AUTO`.

Behavior:

- For ArduPilot vehicles, upload is allowed during an active mission.
- After successful upload, the GCS sends `DO_SET_MISSION_CURRENT` for the active waypoint.
- ArduPilot reloads the current navigation command from the updated mission.
- The aircraft remains in `AUTO`; no manual mode cycling is required.

How to use: 1. Fly the vehicle in `AUTO`. 2. Open **Plan View**. 3. Edit or add waypoints. 4. Click **Upload**. 5. Confirm upload success.

Safety behavior: - If upload fails, no auto-apply command is sent. - If the vehicle is disarmed, not ArduPilot, or not in `AUTO`, auto-apply is skipped. - If the active waypoint index is invalid after upload, auto-apply is skipped.

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7. Fly View – Flight Operations

7.1 Fly View Layout

Fly View displays:

- Map
- Vehicle position and track
- Telemetry
- Video feed

- Quick parameter panel
- Guided action controls

7.2 Pre-Flight Checks

Before takeoff, verify:

- GPS lock
- Battery level
- Link status
- Flight mode
- Mission loaded, if applicable
- No critical warnings

7.3 Arming and Takeoff

1. Confirm area is clear.
2. Select the appropriate flight mode.
3. Arm the vehicle.
4. Initiate takeoff according to vehicle procedure.

7.4 Monitoring During Flight

Monitor:

- Attitude and heading
- Altitude AGL
- Airspeed
- Ground speed
- Flight mode
- Arming status
- Battery status
- GPS status
- Link status
- EKF status
- Vibration status
- Distance to home
- Distance to next waypoint

7.5 Fly Here

The map supports guided movement similar to Mission Planner Fly Here.

How to use: 1. Ensure the vehicle is armed and in a supported guided mode. 2. Click the desired location on the map. 3. Confirm the guided action if prompted. 4. The vehicle moves toward the selected point.

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8. Quick Tab

The Quick tab provides live monitoring of flight parameters.

8.1 Default Tiles

Default tiles include:

- `DIST HOME`
- `DIST WP`
- Attitude data
- Altitude
- Airspeed
- Ground speed
- Other selected telemetry

8.2 Adding Parameters

1. Open the **Quick** tab.
2. Right-click anywhere on the panel.
3. Select parameters from the picker.
4. Click an empty tile to assign a parameter.

8.3 Using Quick Tab in Flight

- Keep critical telemetry visible.
- Use it to monitor distance to home and next waypoint.
- Add parameters needed for the mission.

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9. Actions Tab – In-Flight Overrides

The Actions tab contains Mission Planner–style quick actions and in-flight override controls.

9.1 Change Alt

What it does: Changes the target altitude of the current navigation waypoint without changing flight mode.

- Input is an **absolute target altitude relative to home**.
- Example: entering `100` means target 100 units above home.
- Range: `0` to `10000`.
- Uses Mission Planner's altitude-only `MISSION_ITEM` method:
- `seq = 0`
- `current = 3`
- `MAV_CMD_NAV_WAYPOINT`

- `MAV_FRAME_GLOBAL_RELATIVE_ALT`
- zero latitude/longitude

How to use: 1. Open the **Actions** tab. 2. Enter the target altitude. 3. Click **Set**. 4. Hold the confirmation button for about 1.5 seconds.

9.2 Change Speed

What it does: Changes the target airspeed.

- Range: `0` to `1000`.
- Default value is taken from:
- `WP_SPEED_MAX`, if available
- Otherwise `TRIM_ARSPD_CM`
- Otherwise current airspeed
- Uses Mission Planner's exact `DO_CHANGE_SPEED` parameters:
- `param1 = 0` (airspeed)
- `param2 = target speed`
- remaining parameters = `0`

Important: ArduPilot accepts this only in `AUTO`, `GUIDED`, or `AVOID_ADSB`. In other modes it may be silently ignored.

How to use: 1. Open the **Actions** tab. 2. Enter the target speed. 3. Click **Set**. 4. Hold to confirm.

9.3 Set Loiter Rad

What it does: Sets the loiter radius parameter.

- Range: `-10000` to `10000`.
- Negative values mean counter-clockwise.
- Writes `LOITER_RAD` if present, otherwise `WP_LOITER_RAD`.
- On ArduPlane this is normally `WP_LOITER_RAD`.

How to use: 1. Open the **Actions** tab. 2. Enter the desired radius. 3. Click **Set**. 4. Hold to confirm.

9.4 RTL Radius

RTL radius is not one of Mission Planner's three Actions-tab controls. It remains available through the **Parameters** tab.

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10. Parameters, Save Confirmation, and Password Gate

10.1 Parameter Access

1. Open **Parameters**.
2. Enter the configured password.
3. Browse or search parameters.
4. Modify values as needed.

10.2 Save Confirmation

- When parameters are modified, the GCS prompts for confirmation before applying changes.
- This prevents accidental parameter writes.

10.3 Safety Notes

- Only modify parameters if you understand their effect.
- Confirm changes carefully.
- Keep parameter backups when possible.

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11. Recommended Pilot Workflow

Before Flight

1. Connect vehicle and verify telemetry.
2. Verify video feed.
3. Complete calibrations.
4. Load and verify mission.
5. Upload mission.
6. Confirm parameters and failsafes.

During Flight

1. Monitor Fly View telemetry.
2. Use Quick tab for key data.
3. Use Actions tab for in-flight overrides.
4. Use map guidance if required.
5. Upload updated missions during AUTO if needed.

After Flight

1. Review mission execution.
2. Save logs if required.

3. Inspect any warnings or errors.

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12. Safety Notes

- Change Speed may be ignored outside `AUTO`, `GUIDED`, or `AVOID_ADSB`.
- Negative loiter radius means counter-clockwise.
- Auto-apply mission upload is intended for ArduPilot AUTO missions.
- Fast compass calibration is only suitable where the magnetic field is close to spherical.
- Always validate mission edits, uploads, and calibration changes in a safe test environment before operational use.
- Do not upload unverified missions during critical flight phases.

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13. Summary of FWD LM Custom Features

- Mission Planner Actions tab parity
- Change Alt using altitude-only mission item
- Change Speed using MP-compatible `DO_CHANGE_SPEED`
- Set Loiter Rad using `LOITER_RAD` / `WP_LOITER_RAD`
- Quick tab with live telemetry monitoring
- Fly Here map guidance
- Auto-apply mission upload during AUTO
- Compass calibration and priority support
- Parameter save confirmation and password gate
- FWD LM UI branding and fixed-wing iconography